

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Ratios, rates, proportional relationships, and units

02

10 questions · ~15 min

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

x	y
1	4
3	12
5	20
40	k

In the table above, the ratio of y to x for each ordered pair is constant. What is the value of k ?

- A. 28
- B. 36
- C. 80
- D. 160

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

For a person m miles from a flash of lightning, the length of the time interval from the moment the person sees the lightning to the moment the person hears the thunder is k seconds. The ratio of m to k can be estimated to be 1 to 5. According to this estimate, the person is how many miles from a flash of lightning if the time interval is 25 seconds?

- A. 10
- B. 9
- C. 6
- D. 5

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

How many teaspoons are equivalent to 44 tablespoons? 3 teaspoons = 1 tablespoon

- A. 47
- B. 88
- C. 132
- D. 176

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

An object travels at a constant speed of 6 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 24 centimeters?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

How many meters are equivalent to 2,300 centimeters? 100 centimeters = 1 meter

- A. 0.043
- B. 23
- C. 2,400
- D. 230,000

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

Tilly earns p dollars for every w hours of work. Which expression represents the amount of money, in dollars, Tilly earns for $39w$ hours of work?

- A. $39p$
- B. $\frac{p}{39}$
- C. $p + 39$
- D. $p - 39$

Q8

PROBLEM-SOLVING AND DATA ANALYSIS

A cable provider wanted to know how many of its 30,000 customers would be interested in a new service plan. The provider selected 300 customers at random and asked each customer whether the customer would be interested in the new plan. Of those surveyed, 8 said they would be interested. Which of the following is the best estimate of the total number of customers who would be interested in the new service plan?

- A. 8
- B. 80
- C. 800
- D. 8,000

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

The ratio of t to u is 1 to 2, and $t = 10$. What is the value of u ?

- A. 2
- B. 5
- C. 10
- D. 20

A certain bird species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this bird species fly in 4 seconds?

- A. 64
- B. 20
- C. 16
- D. 12